

Copulas and financial applications (Suite)

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1 Measures of dependence

The **traditional way** of talking about “dependencies”

...but probably not the most relevant one!

For any two r.v. X and Y , with means μ_X and μ_Y , their linear correlation coefficient is

$$\rho(X, Y) := \frac{\text{Cov}(X, Y)}{\sigma(Y)\sigma(X)} = \frac{\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]}{\sqrt{\mathbb{E}[(X - \mu_X)^2]\mathbb{E}[(Y - \mu_Y)^2]}}.$$

Strongly linked to **linear regressions**:

$$(a^*, b^*) := \min_{a, b} \mathbb{E}[(Y - a - bX)^2],$$

$$b^* = \rho(X, Y) \frac{\sigma(Y)}{\sigma(X)}, \quad a^* = \mu_Y - b^* \mu_X.$$

- $|\rho(X, Y)| \leq 1$
- $\rho(X, Y) \in \{1, -1\}$ implies $Y = aX + b$ a.s. for some constants a and b .
- X and Y are independent $\Rightarrow \rho(X, Y) = 0$ but **the opposite is false** in general: $Y = X^2$, X symmetrical, $\mathbb{E}[X] = 0$.

Particular case: (X, Y) is a Gaussian random vector ($\neq$ “ X and Y are Gaussian”!). Then $\rho(X, Y) = 0$ implies $X \perp\!\!\!\perp Y$.

This is not true for Elliptical distributions:

$(X, Y) = S \mathbf{Z}$, $\mathbf{Z} \sim \mathcal{N}(0, Id_2)$, $S \perp\!\!\!\perp \mathbf{Z}$, and then

$$\mathbb{E}[X Y] = \mathbb{E}[S^2 Z_1 Z_2] = \mathbb{E}[S^2] \mathbb{E}[Z_1 Z_2] = 0.$$

- ρ is not invariant wrt to nonlinear monotonic transforms of the margins.
- Necessary to have finite second order moments!

Not always relevant, especially in finance/insurance (fat tails).

Ex.: $\mathbb{P}(|X| > t) \propto t^{-\alpha}$, $\alpha \in (0, 2)$; Student laws with ν degrees of freedom, $\nu < 2$; α -stable laws, etc.

A **more annoying** feature: consider two r.v. X and Y , $\sigma(X) + \sigma(Y) < +\infty$. Their laws are given.

The set of correlations $\rho(X, Y)$ that can be reached is an interval $\mathcal{I}_{X,Y}$, but not $[-1, 1]$ as a whole!

Proposition 1

$\mathcal{I}_{X,Y} = [\rho_{\min}, \rho_{\max}] \subset [-1, 1]$, with $\rho_{\min} \leq 0$ and $\rho_{\max} \geq 0$.

$\rho_{\min}$ (resp. $\rho_{\max}$) is the correlation level when X and Y are counter-monotone (resp. comonotone).

Lemma 1 (Hoeffding's identity)

If $X \sim F$, $Y \sim G$ and $(X, Y) \sim H$, then

$$\text{Cov}(X, Y) = \int \{H(x, y) - F(x)G(y)\} dx dy.$$

Proof of Proposition 1: Fréchet bounds: for every x, y ,

$$\max(F(x) + G(y) - 1, 0) \leq H(x, y) \leq \min(F(x), G(y))$$

$\Rightarrow \mathcal{I}_{X,Y}$ is an interval, whose bounds are reached with C^+ and C^- .

$\rho_{\max} \geq 0$ because $\min(F(x), G(y)) \geq F(x)G(y)$. $\rho_{\min} \leq 0$

because $\max(F(x) + G(y) - 1, 0) \leq F(x)G(y)$. Indeed,

$$(u + v - 1)^+ \leq uv \Leftrightarrow (1 - u)(1 - v) \geq 0.$$

By linear correlation of C^+ and C^- , the whole interval $\mathcal{I}_{X,Y}$ is covered. $\square$

Proof of Lemma 1: a. If (X_1, Y_1) and (X_2, Y_2) are two independent copies of (X, Y) , then

$$2\text{Cov}(X, Y) = \mathbb{E}[(X_1 - X_2)(Y_1 - Y_2)] \text{ (easy).}$$

b. Note that, for all x_1 and x_2 ,

$$x_1 - x_2 = \int_{x_2}^{x_1} du = \int \{1(x_2 \leq u) - 1(x_1 \leq u)\} du.$$

Apply this rule for the calculation of $\mathbb{E}[(X_1 - X_2)(Y_1 - Y_2)]$.

Proof of Lemma 1 (cont'd):

$$\begin{aligned} 2\text{Cov}(X, Y) &= \int \mathbb{E} \left[\{1(X_2 \leq u) - 1(X_1 \leq u)\} \{1(Y_2 \leq v) - 1(Y_1 \leq v)\} \right] du dv \\ &= \int \left\{ \mathbb{E}[1(X_2 \leq u, Y_2 \leq v)] + \mathbb{E}[1(X_1 \leq u, Y_1 \leq v)] \right. \\ &\quad \left. - \mathbb{E}[1(X_2 \leq u, Y_1 \leq v)] - \mathbb{E}[1(X_1 \leq u, Y_2 \leq v)] \right\} du dv \\ &= \int \{2H(u, v) - 2F(u)G(v)\} du dv. \quad \square. \end{aligned}$$

Example: Let Z_1 and Z_2 be two $\mathcal{N}(0, 1)$ r.v.

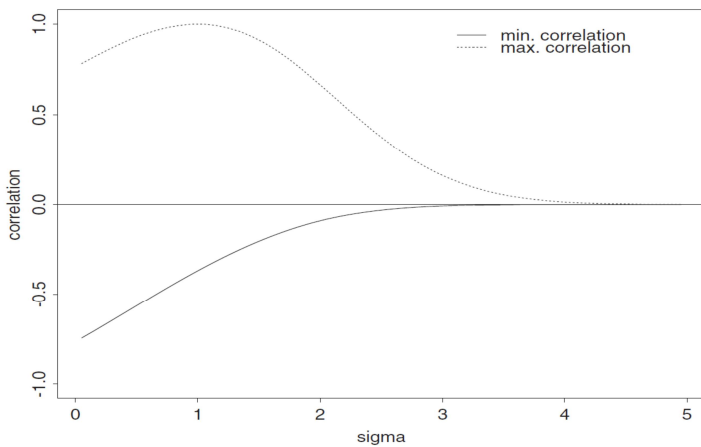
Set $X := \exp(Z_1)$, $Y := \exp(\sigma Z_2)$.

$$\Rightarrow \rho_{\min} = \frac{\exp(-\sigma) - 1}{\sqrt{(e - 1)(\exp(\sigma^2) - 1)}}, \rho_{\max} = \frac{\exp(\sigma) - 1}{\sqrt{(e - 1)(\exp(\sigma^2) - 1)}}.$$

In particular, $\lim_{\sigma \rightarrow \infty} \rho_{\min} = \lim_{\sigma \rightarrow \infty} \rho_{\max} = 0$,

For large values of σ , roughly the same level of correlation when X and Y are comonotone or counter-monotone!

Range of linear correlations as a function of σ



Most of the latter properties of linear correlations are **not in favor of this measure**, at least to be understood as a “dependence measure”.

Nonetheless, covariances/correlations yield natural tools for risk management when the returns in the portfolio are **elliptically distributed**.

Assume $\mathbf{X} \sim \mathcal{E}_m(\mu, \Sigma, g)$ a vector of asset returns, and impose $\text{Cov}(\mathbf{X}) = \Sigma$. A linear portfolio return is $Z := \omega' \mathbf{X}$, and ω is a vector of weights (allocations).

$\Rightarrow Z \sim \mathcal{E}_m(\omega' \mu, \omega' \Sigma \omega, g)$. Its variance is $\omega' \Sigma \omega$, as for Gaussian vectors.

Denote by $\text{VaR}_\alpha(Z)$ the value-at-risk associated to any r.v. Z , for some level $\alpha \in (0, 1)$, and $\mathcal{P}$ denotes the set of portfolio returns Z .

Theorem 1 (Embrechts, McNeil, Straumann, 2002.)

- (i) *The value-at-risk is subadditive in $\mathcal{P}$: for every Z_1 and Z_2 in $\mathcal{P}$: $VaR_\alpha(Z_1 + Z_2) \leq VaR_\alpha(Z_1) + VaR_\alpha(Z_2)$.*
- (ii) *Let ν be is a positively homogeneous risk measure, i.e. a map from $\mathcal{P}$ to $\mathbb{R}$ s.t. $\nu(\lambda Z) = \lambda \nu(Z)$, $\forall \lambda \in \mathbb{R}_+$ and $Z \in \mathcal{P}$: $\nu(Z_1 - \mathbb{E}[Z_1]) \leq \nu(Z_2 - \mathbb{E}[Z_2]) \Leftrightarrow \sigma^2(Z_1) \leq \sigma^2(Z_2)$.*
- (iii) *If ν is a positively homogeneous and translation invariant ($\nu(Z + a) = \nu(Z) + a$, $\forall (a, Z) \in \mathbb{R} \times \mathcal{P}$, then, for any real r ,*

$$\arg \min_{Z \in \tilde{\mathcal{P}}_r} \nu(Z) = \arg \min_{Z \in \tilde{\mathcal{P}}_r} \sigma^2(Z),$$

where $\tilde{\mathcal{P}} = \{Z = \omega' \mathbf{X} \in \mathcal{P}, \sum_{k=1}^m \omega_k = 1, \mathbb{E}[Z] = r\}$.

Dependencies between several random variables = their copula C

Ok, but discussing a m -dimensional map is not very easy.

$\Rightarrow$ a need for a “summary” of information, i.e. a single or a few numbers that depend on C only.

A measure of association between $\kappa(X, Y)$ two random variables is a real number that is defined by their copula C only

Notation: $\kappa(X, Y) := \kappa(C)$.

Definition 1

A numeric measure of association is a *measure of concordance* if

- κ is defined for every continuous r.v. X and Y ;
- $-1 \leq \kappa \leq 1$, $\kappa(X, X) = 1$, $\kappa(X, -X) = (-1)$;
- $\kappa(X, Y) = \kappa(Y, X)$;
- If $X \perp\!\!\!\perp Y$ then $\kappa(X, Y) = 0$;
- $\kappa(X, -Y) = \kappa(-X, Y) = -\kappa(X, Y)$;
- For two copulas C_1 and C_2 , if $C_1 \leq C_2$ then $\kappa(C_1) \leq \kappa(C_2)$;
- If (C_n) converges pointwise to C , then $\kappa(C_n) \rightarrow \kappa(C)$.

$X \sim F$, $Y \sim G$ and C is the copula of (X, Y) .

Definition 2

*Let (X_1, Y_1) and (X_2, Y_2) be two independent versions of (X, Y) .
The Kendall's tau between X and Y is*

$$\tau(X, Y) := \mathbb{P}((X_1 - X_2)(Y_1 - Y_2) > 0) - \mathbb{P}((X_1 - X_2)(Y_1 - Y_2) < 0).$$

Kendall's tau = probability of concordant pairs (of observations) -
probability of discordant pairs (of observations).

Definition 3

Let (X_1, Y_1) , (X_2, Y_2) and (X_3, Y_3) be three independent versions of (X, Y) . The Spearman's rho between X and Y is

$$\rho_S(X, Y) := 3\{\mathbb{P}((X_1 - X_2)(Y_1 - Y_3) > 0) \\ - \mathbb{P}((X_1 - X_2)(Y_1 - Y_3) < 0)\}.$$

Let $(X_0, Y_0) \sim C^\perp$, a vector of two independent components.

Spearman rho = probability of concordant pairs between (X, Y)

and (X_0, Y_0) - probability of discordant pairs between (X, Y) and (X_0, Y_0) .

Kendall's tau and Spearman's rho

Kendall's tau and Spearman's rho depend on the copula only.

Theorem 2

If the laws F and G of X and Y resp. are continuous, then

$$\tau(X, Y) = 4 \int C(u, v) C(du, dv) - 1, \quad (1)$$

$$\rho_S(X, Y) = 12 \int \{C(u, v) - uv\} du dv. \quad (2)$$

Remark: $\tau(X, Y) = 4\mathbb{E}[C(U, V)] - 1$, $(U, V) \sim C$, and $\rho(X, Y) = 12\mathbb{E}[\{C(U_0, V_0) - U_0 V_0\}]$, $(U_0, V_0) \sim C^\perp$.

Corollary 1

If the laws of X and Y are continuous, then their Spearman's rho is the linear correlation between $F(X)$ and $G(Y)$.

Proof of Theorem 2.

Proof of (1): Since the laws are continuous,

$$\mathbb{P}((X_1 - X_2)(Y_1 - Y_2) > 0) = 1 - \mathbb{P}((X_1 - X_2)(Y_1 - Y_2) < 0).$$

$$\begin{aligned}\tau(X, Y) &= 2 \mathbb{P}((X_1 - X_2)(Y_1 - Y_2) > 0) - 1 \\ &= 2(\mathbb{P}(X_1 > X_2, Y_1 > Y_2) + \mathbb{P}(X_1 < X_2, Y_1 < Y_2)) - 1.\end{aligned}$$

$$\begin{aligned}\mathbb{P}(X_1 > X_2, Y_1 > Y_2) &= \mathbb{P}(F(X_1) > F(X_2), G(Y_1) > G(Y_2)) \\ &= \int \mathbb{P}(F(X_2) < u, G(Y_2) < v) C(du, dv) \\ &= \int C(u, v) C(du, dv).\end{aligned}$$

Similarly: $\mathbb{P}(X_1 < X_2, Y_1 < Y_2) = \int C(u, v) C(du, dv).$

Proof of (2):

Note that $(X_1, Y_1) \sim C$ and $(X_2, Y_3) \sim C^\perp$.

$$\begin{aligned}\rho_S(X, Y)/3 &= 2\mathbb{P}((X_1 - X_2)(Y_1 - Y_3) > 0) - 1 \\ &= 2(\mathbb{P}(X_1 > X_2, Y_1 > Y_3) + \mathbb{P}(X_1 < X_2, Y_1 < Y_3)) - 1.\end{aligned}$$

$$\begin{aligned}\mathbb{P}(X_1 < X_2, Y_1 < Y_3) &= \mathbb{P}(F(X_1) < F(X_2), G(Y_1) < G(Y_3)) \\ &= \int \mathbb{P}(F(X_1) < u, G(Y_1) < v) \, du \, dv \\ &= \int C(u, v) \, du \, dv.\end{aligned}$$

$$\begin{aligned}
\mathbb{P}(X_1 > X_2, Y_1 > Y_3) &= \mathbb{P}(F(X_1) > F(X_2), G(Y_1) > G(Y_3)) \\
&= \int \mathbb{P}(F(X_1) > u, G(Y_1) > v) \, du \, dv \\
&= \int \{1 - u - v + C(u, v)\} \, du \, dv = \int C(u, v) \, du \, dv.
\end{aligned}$$

$$\Rightarrow \rho_S(X, Y)/3 = 4 \int C(u, v) \, du \, dv - 1 = 4 \int \{C(u, v) - uv\} \, du \, dv. \quad \square$$

Corollary 2

If $C_1 \leq C_2$, then $\tau(C_1) \leq \tau(C_2)$ and $\rho_S(C_1) \leq \rho_S(C_2)$.

Proof: For Spearman's rho, this is obvious from Theorem 2. For Kendall's tau?

$$\tau(C_1) \leq \tau(C_2) \Leftrightarrow \mathbb{E}[C_1(U_1, V_1)] \leq \mathbb{E}[C_2(U_2, V_2)],$$

where $(U_1, V_1) \sim C_1$ and $(U_2, V_2) \sim C_2$.

By an integration by parts argument:

$$\begin{aligned}\int C_1 dC_2 &= \int_u \left\{ \int_v C_1(u, v) C_2(du, dv) \right\} \\&= \int_u \left\{ [C_1(u, v) C_2(du, v)]_{v=0}^{v=1} - \int_v C_1(u, dv) C_2(du, v) \right\} \\&= \int_u u du - \int_v \left\{ \int_u C_1(u, dv) C_2(du, v) \right\} \\&= \frac{1}{2} - \int_v \left\{ [C_1(u, dv) C_2(u, v)]_{u=0}^{u=1} - \int_u C_1(du, dv) C_2(u, v) \right\} \\&= \frac{1}{2} - \int v dv + \int C_2 dC_1 = \int C_2 dC_1.\end{aligned}$$

$$\begin{aligned}\Rightarrow \mathbb{E}[C_1(U_1, V_1)] &\leq \mathbb{E}[C_2(U_1, V_1)] = \int C_2 dC_1 \\&= \int C_1 dC_2 \leq \int C_2 dC_2 = \mathbb{E}[C_2(U_2, V_2)]. \quad \square\end{aligned}$$

Proposition 2

Assume F and G are strictly increasing. The following assertions are equivalent:

- (i) $\rho_S(X, Y) = 1$,
- (ii) $\tau(X, Y) = 1$,
- (iii) $C = C^+$ a.s.,
- (iv) $Y = T(X)$ a.s., for an increasing map T .

Kendall's tau and Spearman's rho

Proof of Proposition 2:

We already know that (iii) is equivalent to (iv).

(i) $\Rightarrow$ (iii): If $\rho_S(X, Y) = 1$ then the linear correlation between $F(X)$ and $G(Y)$ is one, and then $\exists(a, b)$, $b > 0$ s.t.

$$F(X) = a + bG(Y), \text{ a.s.}$$

$$\Rightarrow X = F^{-1}(a + bG(Y)) \text{ a.s. and } C = C^+.$$

(iii) $\Rightarrow$ (i): if $C = C^+$, then

$$\int C^+(u, v) du dv = 2 \int u 1(u < v) du dv = \int v^2 dv = \frac{1}{3}$$

Since $\int uv du dv = 1/4$, this yields

$$\rho_S(X, Y) = 12\left(\frac{1}{3} - \frac{1}{4}\right) = 1.$$

Proof of Proposition 2 cont'd:

(iii) $\Rightarrow$ (ii): If $C = C^+$, then $Y = T(X)$ a.s., for some increasing map T .

$$\mathbb{P}((X_1 - X_2)(Y_1 - Y_2) > 0) = \mathbb{P}((X_1 - X_2)(T(X_1) - T(X_2)) > 0) = 1,$$

$$\mathbb{P}((X_1 - X_2)(Y_1 - Y_2) < 0) = \mathbb{P}((X_1 - X_2)(T(X_1) - T(X_2)) < 0) = 0.$$

(ii) $\Rightarrow$ (iii): Let $(U, V) \sim C$ and $(U^+, V^+) \sim C^+$

$$\begin{aligned}\mathbb{E}[C(U, V)] &\leq \mathbb{E}[C^+(U, V)] = \mathbb{E}[C(U^+, V^+)] \\ &\leq \mathbb{E}[C^+(U^+, V^+)].\end{aligned}$$

If $\tau(X, Y) = 1$ then $\mathbb{E}[C(U, V)] = \mathbb{E}[C^+(U^+, V^+)]$ (Th. 2).

Proof of Proposition 2 Cont'd:

$$\Rightarrow \mathbb{E}[C(U, V)] = \mathbb{E}[C^+(U, V)],$$

and $C = C^+$, μ_C -a.s.

Similarly, $\mathbb{E}[C(U^+, V^+)] = \mathbb{E}[C^+(U^+, V^+)]$ and $C = C^+$, μ_{C^+} -a.s.

Set $A := \{(u, v) | C(u, v) \neq C^+(u, v)\}$, then $\mu_C(A) = \mu_{C^+}(A) = 0$.

For every box B ,

$$\begin{aligned} C(B) &= C(B \cap A^c) + C(B \cap A) \\ &= C^+(B \cap A^c) + 0 \\ &= C^+(B) - C^+(B \cap A) = C^+(B) \end{aligned}$$

$\Rightarrow C = C^+$ a.s. $\square$

Proposition 3

Assume F and G are strictly increasing. The following assertions are equivalent:

- (i) $\rho_S(X, Y) = (-1)$,
- (ii) $\tau(X, Y) = (-1)$,
- (iii) $C = C^-$ a.s.,
- (iv) $Y = T(X)$ a.s., for a decreasing map T .

$\Rightarrow$ Kendall's tau and Spearman's rho are **measures of concordance**.

Let $(X, Y) \sim (C, F, G)$ and we observe a sample $((X_i, Y_i))_{i=1, \dots, n}$.

$$\hat{F}(t) = n^{-1} \sum_{i=1}^n 1(X_i \leq t), \quad \hat{G}(t) = n^{-1} \sum_{i=1}^n 1(Y_i \leq t)$$

$$\hat{U}_i := \hat{F}(X_i), \quad \hat{V}_i := \hat{G}(Y_i), \quad \forall i.$$

$\hat{\rho}_S(X, Y)$ is defined as the empirical correlation between the $\hat{U}_i$ and the $\hat{V}_i$, $i \in \{1, \dots, n\}$.

$$\hat{\rho}_S(X, Y) \rightarrow \rho_S(X, Y) \text{ a.s.}$$

Kendall's tau and Spearman's rho: inference

Consider the $n(n-1)/2$ pairs of observations among $\{(X_i, Y_i); i = 1, \dots, n\}$.

c_n = number of concordant pairs, i.e. the pairs (i, j) s.t. $(X_i - X_j)(Y_i - Y_j) > 0$.

d_n = number of discordant pairs, i.e. the pairs (i, j) s.t. $(X_i - X_j)(Y_i - Y_j) < 0$.

$$\hat{\tau}(X, Y) := \frac{c_n - d_n}{c_n + d_n} \rightarrow \tau(X, Y) \text{ a.s.}$$

Proposition 4

Let $(X, Y) \sim (C, F, G)$ and C is Archimedean with a strict generator h , $h' < 0$. Then, $\tau(X, Y) = 1 + 4 \int_0^1 h(t)/h'(t) dt$.

Proof of Prop. 4: Let $U := F(X)$ and $V := G(Y)$. Define the **Kendall function** K of C as the cdf of $C(U, V)$. We will prove that

$$K(t) = t - h(t)/h'(t), \quad t \in (0, 1). \quad (3)$$

$$\begin{aligned} \Rightarrow 4\mathbb{E}[C(U, V)] - 1 &= 4 \int_0^1 t K(dt) - 1 \\ &= 4 \left\{ [tK(t)]_0^1 - \int_0^1 K(t) dt \right\} - 1 \\ &= 3 - 4 \int_0^1 K(t) dt = 1 + 4 \int_0^1 \frac{h(t)}{h'(t)} dt. \end{aligned}$$

Proof of Equation (3): for every $t \in (0, 1)$,

$$\begin{aligned}
 \mathbb{P}(C(U, V) \leq t) &= \mathbb{P}(h^{-1}(h(U) + h(V)) \leq t) \\
 &= \mathbb{P}(h(U) + h(V) \geq h(t)) \\
 &= \mathbb{P}\left(U \leq h^{-1}(h(t) - h(V)); h(t) \geq h(V)\right) \\
 &\quad + \mathbb{E}[1(h(t) < h(V))].
 \end{aligned}$$

Since the density of $U|V = v$ is $\partial_2 C(\cdot, v)$, we have

$$\begin{aligned}
 \mathbb{P}(C(U, V) \leq t) &= \mathbb{E}\left[\partial_2 C(h^{-1}(h(t) - h(V)), V) 1(h(t) \geq h(V))\right] \\
 &\quad + \mathbb{P}(V < t) \\
 &= \mathbb{E}\left[\partial_2 C(h^{-1}(h(t) - h(V)), V) 1(t \leq V)\right] + \mathbb{P}(V < t).
 \end{aligned}$$

Proof of Equation (3) cont'd: Note that

$$\partial_2 C(u, v) = \frac{h'(v)}{h' \circ h^{-1}(h(u) + h(v))}$$

$$\Rightarrow \partial_2 C(h^{-1}(h(t) - h(v)), v) = \frac{h'(v)}{h'(t)}$$

$$\begin{aligned} \mathbb{P}(C(U, V) \leq t) &= \mathbb{E}\left[\frac{h'(V)}{h'(t)} 1(t \leq V)\right] + t \\ &= \frac{h(1) - h(t)}{h'(t)} + t = t - \frac{h(t)}{h'(t)}. \quad \square \end{aligned}$$

If the copula of (X, Y) is elliptical, with a correlation parameter $\rho \in [-1, 1]$, then

$$\tau(X, Y) = \frac{2}{\pi} \arcsin(\rho).$$

If the copula of (X, Y) is Gaussian with a correlation parameter ρ , then

$$\rho_S(X, Y) = \text{Corr}(\Phi(X), \Phi(Y)) = \frac{6}{\pi} \arcsin\left(\frac{\rho}{2}\right).$$

$\Rightarrow$ two simple and nice ways of estimating Gaussian copulas!

- the Blomqvist coefficient

$$\rho_B := 4\mathbb{P}(U \leq 1/2, V \leq 1/2) - 1 \in [-1, 1].$$

- the Gini coefficient

$$\gamma := 2\mathbb{E}[|U+V-1| - |U-V|] = 2 \int \{|u+v-1| - |u-v|\} C(du, dv).$$

- Let $(X, Y) \sim (C, F, G)$, and set $Z_1 := \Phi^{-1} \circ F(X)$ and $Z_2 := \Phi^{-1} \circ G(Y)$. The van der Waerden coefficient of (X, Y) is $\rho_W := \text{Corr}(Z_1, Z_2)$.

Note that Z_k is $\mathcal{N}(0, 1)$, $k = 1, 2$, but (Z_1, Z_2) is not a Gaussian random vector in general.

In many situations, an interest in analyzing joint tail events and their probability

$\Rightarrow$ we need indicators of “joint strong moves”. They should not depend on marginal laws (large move $\simeq$ related to high quantiles)

$\Rightarrow$ when $(X, Y) \sim (C, F, G)$, some tail dependence indicators of (X, Y) have been proposed. And they depend on the underlying copula C only.

To simplify, we assume the law of (X, Y) is continuous:
 $U := F(X)$ and $V := G(Y)$ are $\mathcal{U}([0, 1])$.

Definition 4

The lower tail dependence parameter of (X, Y) is

$$\begin{aligned}\lambda_L &:= \lim_{u \rightarrow 0^+} \mathbb{P}(Y < G^-(u) | X < F^-(u)) \\ &= \lim_{u \rightarrow 0^+} \mathbb{P}(X < F^-(u) | Y < G^-(u)) \\ &= \lim_{u \rightarrow 0^+} C(u, u)/u \\ &= \lim_{u \rightarrow 0^+} \frac{\partial}{\partial u} C(u, u) \\ &= \lim_{u \rightarrow 0^+} \{ \mathbb{P}(U \leq u | V = u) + \mathbb{P}(V \leq u | U = u) \}.\end{aligned}$$

Denote by $S_C(u, v) = \mathbb{P}(F(X) > u, G(Y) > v)$ the bivariate survival function of $(F(X), G(Y))$.

Definition 5

The upper tail dependence parameter of (X, Y) is

$$\begin{aligned}\lambda_U &:= \lim_{u \rightarrow 1^-} \mathbb{P}(Y > G^-(u) | X > F^-(u)) \\ &= \lim_{u \rightarrow 1^-} \mathbb{P}(X > F^-(u) | Y > G^-(u)) \\ &= \lim_{u \rightarrow 1^-} S_C(u, u) / (1 - u) \\ &= - \lim_{u \rightarrow 1^-} \frac{\partial}{\partial u} S_C(u, u) \\ &= - \lim_{u \rightarrow 1^-} \{ \mathbb{P}(U > u | V = u) + \mathbb{P}(V > u | U = u) \}.\end{aligned}$$

Remark: λ_L and λ_U belong to $[0, 1]$.

Warning: S_C is a survival function, and has not to be confused with the survival copula $\tilde{C}$ of (X, Y) , that is a cdf!

Exercise: prove that $S_C(u, v) = \tilde{C}(1 - u, 1 - v)$, $(u, v) \in [0, 1]^2$.

Definition 6

X and Y are left-tail (resp. right-tail) dependent if $\lambda_L > 0$ (resp. $\lambda_U > 0$). X and Y are asymptotically tail independent if $\lambda_L = 0$ and $\lambda_U = 0$.

Tail dependence: the Gaussian copula case

Assume (U, V) follows a Gaussian copula with a parameter ρ .

$\Rightarrow (\Phi^{-1}(U), \Phi^{-1}(V))$ is a centered bivariate Gaussian vector, with covariance (correlation) ρ :

$$\Phi^{-1}(V) = \rho \Phi^{-1}(U) + \sqrt{1 - \rho^2} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1),$$

and $\varepsilon \perp U$.

$$\Rightarrow \mathbb{P}(V \leq u | U = u) = \Phi\left(\frac{\Phi^{-1}(u) - \rho \Phi^{-1}(u)}{\sqrt{1 - \rho^2}}\right) = \Phi\left(\Phi^{-1}(u) \sqrt{\frac{1 - \rho}{1 + \rho}}\right)$$

Proposition 5

For the Gaussian copula, if $|\rho| < 1$, then $\lambda_L = \lambda_U = 0$.

No tail dependence with Gaussian copulas!

Tail dependence: the Student copula case

Assume (U, V) follows a Student copula with a parameter (ν, ρ) .

(ν, ρ)	-0.5	0	0.5	0.9	1
2	0.06	0.18	0.39	0.72	1
4	0.01	0.08	0.25	0.63	1
10	0.0	0.01	0.08	0.46	1
$+\infty$	0.0	0.0	0.0	0.0	1

Table: Tail dependence parameters λ_U and λ_L for Student copulas with parameters (ρ, ν) .

2 Statistical inference of copula models

Let $\mathbf{X} \in \mathbb{R}^m$, $\mathbf{X} \sim (C, F_1, \dots, F_m)$

To simplify, we assume the underlying law is continuous: C and the F_k are continuous.

We observe n iid realizations of $\mathbf{X}$: a sample $\mathbf{X}_1, \dots, \mathbf{X}_n$,
 $\mathbf{X}_i := (X_{i,1}, \dots, X_{i,m})$.

Q: how can we estimate C ?

Different situations, depending on our assumptions concerning the margins.

Table: Different inference problems with copulas.

		Copula	
		Parametric	Nonparametric
Margins {	Parametric	Full Param	Semi-Param II
	Nonparametric	Semi-Param I	Full Non Param

Assumptions:

- any marginal distribution F_k , $k = 1, \dots, m$ belongs to a parametric family $\mathcal{F}_k := \{F_{k,\theta_k}, \theta_k \in \Theta_k\}$, $\Theta_k \subset \mathbb{R}^{p_k}$;
- the copula C belong to a parametric copula family $\mathcal{F} := \{C_{\theta_c}, \theta_c \in \Theta\}$, $\Theta \in \mathbb{R}^{p_c}$.

$\Rightarrow$ a fully parametric model for the law of $\mathbf{X}$.

Solution 1: “brute-force” inference of the full vector of parameters $\vec{\theta} := (\theta_1, \dots, \theta_m, \theta_c)$.

For instance, by Maximum Likelihood, assuming C and the F_k have densities (c_θ and f_{k,θ_k} resp.) w.r.t. the Lebesgue measure:

$$\arg \min_{\vec{\theta}} L_n(\vec{\theta}) := \sum_{i=1}^n \left\{ \ln c_{\theta_c}(F_{1,\theta_1}(X_{i,1}), \dots, F_{m,\theta_m}(X_{i,m})) \right. \\ \left. + \ln f_{1,\theta_1}(X_{i,1}) + \dots + \ln f_{m,\theta_m}(X_{i,m}) \right\}.$$

Theoretically feasible...but numerically demanding when m increases.

Solution 2: a “two-stage” procedure

- (1) Estimate every marginal model, i.e. estimate θ_k for every $k \in \{1, \dots, m\} \Rightarrow \hat{\theta}_k$, a consistent estimator of θ_k .
- (2) Estimation of the copula parameter θ_c , by plug-in (replace any θ_k by its estimate $\hat{\theta}_k$).

For instance, the second stage can be led by Maximum Likelihood:

$$\hat{\theta}_c := \arg \min_{\theta_c} L_n^{(2)}(\theta_c) := \sum_{i=1}^n \ln c_{\theta_c}(F_{1,\hat{\theta}_1}(X_{i,1}), \dots, F_{m,\hat{\theta}_m}(X_{i,m})).$$

This method is called **Inference For Margins** (IFM).

More practical and realistic than the “brute-force” solution.

Under some conditions of regularity, IFM provides consistent and asymptotically normal estimators, but is not as efficient as the “brute-force” single-stage estimator.

$$\hat{\theta}_c = \arg \max_{\theta_c \in \Theta_c} Q_n(\theta_c, \hat{\theta}_1, \dots, \hat{\theta}_m),$$

for some measurable random function Q_n .

Theorem 3

Assume that (i) Θ_c is compact; (ii) $Q_n(\cdot)$ satisfies a uniform law of large numbers (ULLN) i.e.

$\sup_{\theta_c \in \Theta_c} \sup_{(\theta_1, \dots, \theta_m) \in \mathcal{V}} |Q_n(\vec{\theta}) - Q_\infty(\vec{\theta})| \xrightarrow{P} 0$, where $\mathcal{V}$ denotes a neighborhood of $(\theta_{0,1}, \dots, \theta_{0,m})$; (iii) $\vec{\theta} \mapsto Q_\infty(\vec{\theta})$ is continuous and is uniquely maximized at θ_0 . Then $\hat{\theta}_c$ tends to $\theta_{0,c}$ in probability.

Consider that our (first- and second-stage) estimates are solutions of some first-order conditions (Z-estimators).

The estimator $\hat{\theta}_k$ of θ_k satisfies

$$\int \psi_k(x, \hat{\theta}_k) \mathbb{P}_n(dx) = n^{-1} \sum_{i=1}^n \psi_k(X_{i,k}, \hat{\theta}_k) = 0,$$

for some function ψ_k from $\mathbb{R} \times \Theta_k$ to $\mathbb{R}^{p_k}$.

Typically, imagine we apply the method of moments or the maximum likelihood method, independently for every margin.

The functions ψ_k are given by the derivatives of such criterion w.r.t. θ_k (i.e. the score, in the case of MLE).

The copula parameter $\theta_c \in \Theta_c$ is estimated as a final stage by $\hat{\theta}_c$, that satisfies

$$\int \psi_c(\mathbf{x}, \hat{\theta}_1, \dots, \hat{\theta}_m, \hat{\theta}_c) \mathbb{P}_n(d\mathbf{x}) = n^{-1} \sum_{i=1}^n \psi_c(\mathbf{X}_i, \hat{\theta}) = 0,$$

for some functions ψ_c from $\mathbb{R}^m \times \Theta$ to $\mathbb{R}^{p_c}$, by setting $\hat{\theta} := (\hat{\theta}_1, \dots, \hat{\theta}_m, \hat{\theta}_c)$ and $\Theta := \Theta_1 \times \dots \times \Theta_m \times \Theta_c$.

$$\psi(\mathbf{x}, \vec{\theta}) := \left[\psi_1^T(\mathbf{x}, \theta_1), \dots, \psi_m^T(\mathbf{x}, \theta_m), \psi_c^T(\mathbf{x}, \theta_c) \right]^T,$$

that is defined on $\mathbb{R}^m \times \Theta$.

Theorem 4

Assume that

- (i) $\mathbb{E}[\psi(\mathbf{X}, \theta)] = 0$ has a unique solution θ_0 ;
- (ii) $\hat{\theta}$ tends to θ_0 in probability, and θ_0 belongs to the interior of Θ ;
- (iii) $n^{1/2} \int \psi(\mathbf{x}, \theta_0) \mathbb{P}_n(d\mathbf{x})$ weakly tends to a Gaussian random vector, i.e.

$$n^{-1} \sum_{i=1}^n \psi(\mathbf{X}_i, \theta_0) \rightsquigarrow \mathcal{N}(0, \mathbb{E}[\psi(\mathbf{x}, \theta_0) \psi^T(\mathbf{x}, \theta_0)]);$$

- (iv) $\nabla_{\theta^T} \psi(\mathbf{x}, \theta)$ exists, satisfies a weak ULLN in the neighborhood of θ_0 : $\sup_{\theta \in \mathcal{V}_0} \|\nabla_{\theta^T} \psi(\mathbf{x}, \theta) - \mathbb{E}[\nabla_{\theta^T} \psi(\mathbf{x}, \theta)]\| = o_P(1)$, for a neighborhood $\mathcal{V}_0$ of θ_0 . Moreover, the map $\theta \mapsto \mathbb{E}[\nabla_{\theta^T} \psi(\mathbf{x}, \theta)]$ is continuous and non-singular at θ_0 .

Then $\hat{\theta}$ strongly tends to θ_0 and $n^{1/2}(\hat{\theta} - \theta_0) \rightsquigarrow \mathcal{N}(0, V)$.

$$V := W^{-1}\Sigma(W^{-1})^T, \quad W := \begin{bmatrix} W_{11} & 0 \\ W_{21} & W_{22} \end{bmatrix},$$

$$W_{11} := \text{Diag}\left(\mathbb{E}[\nabla_{\theta_1^T}\psi_1(X_1, \theta_{0,1})], \dots, \mathbb{E}[\nabla_{\theta_d^T}\psi_d(X_m, \theta_{0,m})]\right),$$

$$W_{21} := \mathbb{E}[\nabla_{(\theta_1, \dots, \theta_m)^T}\psi_c(\mathbf{X}, \theta_0)], \quad W_{22} := \mathbb{E}[\nabla_{\theta_c^T}\psi_c(\mathbf{X}, \theta_0)],$$

and Σ is a block-matrix of size $\sum_{k=1}^m p_k + p_c$, whose (i, j) block is

$$\Sigma_{i,j} = \mathbb{E}[\psi_i(\mathbf{X}, \theta_{0,i})\psi_j(\mathbf{X}, \theta_{0,j})^T], \quad i, j = 1, \dots, m,$$

$$\Sigma_{i,m+1} = \mathbb{E}[\psi_i(\mathbf{X}, \theta_{0,i})\psi_c(\mathbf{X}, \theta_0)^T], \quad \Sigma_{m+1,i} = \Sigma_{i,m+1}^T, \quad i \leq m,$$

$$\Sigma_{m+1,m+1} = \mathbb{E}[\psi_c(\mathbf{X}, \theta_0)\psi_c(\mathbf{X}, \theta_0)^T].$$

Nonparametric copula and nonparametric margins

No model assumption $\Rightarrow$ **the empirical copula**.

$F_n :=$ emp. cdf of $(\mathbf{X}_i)_{i=1,\dots,n}$; $F_{n,k} :=$ emp. cdf of $(X_{i,k})_{i=1,\dots,n}$

$$C_n(\mathbf{u}) := F_n(F_{n,1}^{-1}(u_1), \dots, F_{n,m}^{-1}(u_m)), \text{ or}$$

$$\overline{C}_n(\mathbf{u}) := \frac{1}{n} \sum_{i=1}^n 1(F_{n,1}(X_{i,1}) \leq u_1, \dots, F_{n,m}(X_{i,m}) \leq u_m).$$

...but neither C_n or $\overline{C}_n$ are copulas! Actually, there is no significant discrepancy.

Lemma 1

If there are no ties on every marginal sample $(X_{i,k})_{i=1,\dots,n}$, $k = 1, \dots, m$, then $\|C_n - \overline{C}_n\|_\infty \leq mn^{-1}$.

$\Rightarrow$ same asymptotic behaviors of $\sqrt{n}(C_n - C)$ and $\sqrt{n}(\overline{C}_n - C)$.

Reminder: Consider a sequence of probability spaces $(\Omega_n, \mathcal{A}_n, P_n)$ and an sequence of maps $X_n : \Omega_n \rightarrow \mathbb{D}$, a separable metric space (there is a dense countable sequence in A).

Definition 7

(X_n) weakly tends to a random element X that is Borel-measurable if $\mathbb{E}^[f(X_n)] \rightarrow \mathbb{E}[f(X)]$ for every bounded and continuous function $f : \mathbb{D} \rightarrow \mathbb{R}$.*

Particular case of random vectors: $\mathbb{D} = \mathbb{R}^q$: X_n are q -dimensional random vectors; the weak convergence of (X_n) to X is **equivalent to the usual concept of convergence in law**.

The most usual process is the so-called empirical process, associated to an i.i.d. sample $(\mathbf{X}_i)_{i=1,\dots,n}$ of realizations of a random vector $\mathbf{X} \in \mathbb{R}^m$, $\mathbf{X} \sim F$:

$$\mathbb{F}_n = \sqrt{n}(F_n - F), \quad F_n(\mathbf{x}) := \frac{1}{n} \sum_{i=1}^n 1(\mathbf{X}_i \leq \mathbf{x}), \quad \mathbf{x} \in \mathbb{R}^m.$$

$\mathbb{F}_n$ may be seen as a random map from Ω_n to $\mathbb{D} = \ell^\infty(\mathbb{R}^m)$, the space of bounded measurable maps from $\mathbb{R}^m$ to $\mathbb{R}$.

Weak convergence of the empirical copula process

More generally: for any measurable map $g : \mathbb{R}^m \rightarrow \mathbb{R}$, set

$$Pg := \int g dP = \mathbb{E}_{\mathbf{X} \sim P}[g(\mathbf{X})], \quad Fg := \int g dF = \mathbb{E}_{\mathbf{X} \sim F}[g(\mathbf{X})],$$

$$\mathbb{F}_n g := \int g d\mathbb{F}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \{g(\mathbf{X}_i) - \int g dF\}.$$

$\Rightarrow \mathbb{F}_n$ can be seen as random mapping with values in the space of bounded functions $\ell^\infty(\mathcal{T})$, $\mathcal{T}$ being a space of maps, endowed with the usual sup-norm: $\rho(f, g) = \|f - g\|_\infty$ for every bounded real-valued functions f and g .

Weak convergence in $\ell^\infty(A)$, $A \subset \mathbb{R}^m$ = weak convergence in $\ell^\infty(\mathcal{T})$, where $\mathcal{T} := \{g_{\mathbf{y}} : \mathbf{x} \mapsto 1(\mathbf{x} \leq \mathbf{y}); \mathbf{y} \in A\}$.

Weak convergence of the empirical copula process

A m -dimensional Brownian Bridge $\mathbb{B}_F$ defined on $T \subset \mathbb{R}^m$ is a centered Gaussian process whose covariance operator is

$$\mathbb{E}[\mathbb{B}_F(\mathbf{x})\mathbb{B}_F(\mathbf{y})] = F(\mathbf{x} \wedge \mathbf{y}) - F(\mathbf{x})F(\mathbf{y}), \quad \mathbf{x}, \mathbf{y} \in T,$$

by denoting $\mathbf{x} \wedge \mathbf{y} := (\min(x_1, y_1), \dots, \min(x_m, y_m))$.

Theorem 5 (Donsker)

The sequence (F_n) weakly tends to a m -dimensional Brownian bridge $\mathbb{B}_F$ in $\ell^\infty(\mathbb{R}^m)$.

$\Rightarrow$ easy to deduce the law of many statistics (with the continuous mapping theorem and the Functional Delta Method).

Theorem 6 (continuous mapping theorem)

Let $g : \mathbb{D} \rightarrow \mathbb{E}$ be continuous at every point of a set $\mathbb{D}_0 \subset \mathbb{D}$. If $X_n \rightsquigarrow X$ and X takes its values in $\mathbb{D}_0$, then $g(X_n) \rightsquigarrow g(X)$

Let $\mathbb{D}$ and $\mathbb{E}$ be two normed spaces.

Definition 8

A map $\psi : \mathbb{D} \rightarrow \mathbb{E}$ is called Hadamard-differentiable at $\theta \in \mathbb{D}$ tangentially to a set $\mathbb{D}_0 \subset \mathbb{D}$ if there is a continuous linear map $\psi'_\theta : \mathbb{D} \rightarrow \mathbb{E}$ s.t. $\{\psi(\theta + t_n h_n) - \psi(\theta)\} / t_n$ tends to $\psi'_\theta(h)$ when $n \rightarrow \infty$, for any converging sequences $t_n \rightarrow 0$ and $h_n \rightarrow h$, $h \in \mathbb{D}_0$, such that $\theta + t_n h_n \in \mathbb{D}$ for every n .

Proposition 6

If $\phi : \mathbb{D} \rightarrow \mathbb{E}$ is Hadamard-differentiable at θ tangentially to $\mathbb{D}_0$, and if $\psi : \mathbb{E} \rightarrow \mathbb{F}$ is Hadamard-differentiable at $\phi(\theta)$ tangentially to $\phi'_\theta(\mathbb{D}_0)$, then $\psi \circ \phi : \mathbb{D} \rightarrow \mathbb{F}$ is Hadamard-differentiable at θ tangentially to $\mathbb{D}_0$, with derivative $\psi'_{\phi(\theta)} \circ \phi'_\theta$.

Theorem 7 (Functional Delta Method)

Let $\mathbb{D}$ and $\mathbb{E}$ be two metrizable topological vector spaces. Let $\phi : \mathbb{D} \rightarrow \mathbb{E}$ be Hadamard-differentiable at θ tangentially to $\mathbb{D}_0$. Let $X_n : \Omega_n \rightarrow \mathbb{D}$ be maps with $r_n(X_n - \theta) \rightsquigarrow X$ for some sequence of constants $r_n \rightarrow \infty$, where X is separable^a and takes its values in $\mathbb{D}_0$. Then $r_n(\phi(X_n) - \phi(\theta)) \rightsquigarrow \phi'_\theta(X)$. If ϕ'_θ is defined and continuous on the whole of $\mathbb{D}$, then the sequence $r_n(\phi(X_n) - \phi(\theta)) - \phi'_\theta(r_n(X_n - \theta))$ converges to zero in (outer) probability.

^ai.e. there is a measurable subset A s.t. $\mathbb{P}(X \in A) = 1$ and A is separable.

Weak convergence of the empirical copula process

Let $\mathbb{C}_n := \sqrt{n}(C_n - C)$ be the empirical copula process.

Let f be any measurable bounded map from $[0, 1]^m$ to $\mathbb{R}$.

$(\mathbb{C}_n)$ is a process on $\ell^\infty([0, 1]^m)$: $\mathbb{C}_n f = \int f d\mathbb{C}_n$.

$\mathbf{X} \in \mathbb{R}^m$, $\mathbf{X} \sim H$. Note that the mapping H into its copula function C can be decomposed as

$$\phi : H \mapsto (H, H_1, \dots, H_d) \mapsto (H, H_1^{-1}, \dots, H_d^{-1}) \mapsto H(H_1^{-1}, \dots, H_m^{-1}).$$

$\Rightarrow$ we can prove that the map $H \mapsto H \circ (H_1^{-1}, \dots, H_m^{-1}) =: C$ is Hadamard differentiable.

Denote $D_m([0, 1]^m)$ is the space of real-valued “cadlag” functions on $[0, 1]^m$ and $C([0, 1]^m)$ the space of continuous functions on $[0, 1]^m$.

Lemma 2

Let H a cdf that is continuously differentiable on its compact support $[0, 1]^m$. Its marginal distributions H_k , $k = 1, \dots, m$, are C^1 on their supports, with strictly positive densities h_k . Then, the map $\phi : D_m([0, 1]^m) \rightarrow \ell^\infty([0, 1]^m)$ defined by,
 $\phi(H)(\mathbf{u}) = H(H_1^{-1}(u_1), \dots, H_m^{-1}(u_m))$, $\mathbf{u} \in [0, 1]^m$, which transforms the cdf H into its copula function C , is Hadamard differentiable tangentially to $C([0, 1]^m)$. Setting $\mathbf{x} := (H_1^{-1}(u_1), \dots, H_m^{-1}(u_m))$, its derivative at H is

$$\phi'_H(g)(\mathbf{u}) = g(\mathbf{x}) - \sum_{k=1}^m \frac{\partial H}{\partial x_k}(\mathbf{x}) \frac{g(H_k^{-1}(u_k), \infty_{-k})}{h_k(H_k^{-1}(u_k))}.$$

Weak convergence of the empirical copula process

Let $\mathbb{B}_C$ be a m -dimensional Brownian bridge, i.e. a Gaussian random process whose covariance function is given by

$$\mathbb{E}[\mathbb{B}_C(\mathbf{u})\mathbb{B}_C(\mathbf{v})] = C(\mathbf{u} \wedge \mathbf{v}) - C(\mathbf{u})C(\mathbf{v}), \quad \mathbf{u}, \mathbf{v} \in [0, 1]^m.$$

For every $j \in \{1, \dots, m\}$, the j -th partial derivative of C is denoted by $\dot{C}_j$, when it exists.

$$\mathbb{C}(\mathbf{u}) := \mathbb{B}_C(\mathbf{u}) - \sum_{k=1}^m \dot{C}_k(\mathbf{u})\mathbb{B}_C(1_k, u_k), \quad \mathbf{u} \in [0, 1]^m.$$

Th. 7+ Lemma 2 imply:

Theorem 8

Assume the law of $\mathbf{X}$ is continuous, and all the partial derivatives $\dot{C}_k$ exist and are continuous on the whole hypercube $[0, 1]^m$, $k = 1, \dots, m$. Then, $\mathbb{C}_n \rightsquigarrow \mathbb{C}$ in $\ell^\infty([0, 1]^m)$.

Unfortunately, many usual copulas are not continuously differentiable on the whole hypercube $[0, 1]^m$.

For instance, for the Gaussian copula with nonzero correlation parameter ρ , the first-order partial derivatives fail to be continuous at the points $(0, 0)$ and $(1, 1)$ if $\rho > 0$ and at the points $(0, 1)$ and $(1, 0)$ if $\rho < 0$.

Segers (2012) has weakened the regularity assumption of Theorem 8.

Assumption 1

For every $j \in \{1, \dots, m\}$, the j -th partial derivative of C , denoted by $\dot{C}_j$, exists and is continuous on the set $V_j := \{\mathbf{u} \in [0, 1]^m, 0 < u_j < 1\}$. We extend the domain of $\dot{C}_j$ to the whole of $[0, 1]^m$ by setting $\dot{C}_j(\mathbf{u}) = \limsup_{h \rightarrow 0^+} C(\mathbf{u} + h\mathbf{e}_j)/h$ if $u_j = 0$, and $\dot{C}_j(\mathbf{u}) = \limsup_{h \rightarrow 0^+} (C(\mathbf{u}) - C(\mathbf{u} - h\mathbf{e}_j))/h$ if $u_j = 1$.

Theorem 9 (Segers, 2012)

Under Assumption 1, $\mathbb{C}_n \rightsquigarrow \mathbb{C}$ in $\ell^\infty([0, 1]^m)$.

Application: Goodness-of-fit test.

Let C_0 be a given copula. We want to test

$$\mathcal{H}_0 : C = C_0, \text{ against } \mathcal{H}_a : C \neq C_0.$$

$$\text{Test statistics: } T_n^{(KS)} := \sup_{\mathbf{u} \in [0,1]^m} \sqrt{n} |(C_n - C_0)(\mathbf{u})|,$$

$$T_n^{(CvM)} := n \int_{\mathbf{u} \in [0,1]^m} (C_n - C_0)^2(\mathbf{u}) d\mathbf{u},$$

$$T_n^{(CvM2)} := n \int_{\mathbf{u} \in [0,1]^m} (C_n - C_0)^2(\mathbf{u}) C_0(d\mathbf{u}), \text{ etc.}$$

Under $\mathcal{H}_0$, $T_n^{(KS)}$ (resp. $T_n^{(CvM)}$) tends in law to

$$T_\infty^{(KS)} := \sup_{\mathbf{u} \in [0,1]^m} |\mathbb{C}(\mathbf{u})| \text{ (resp. } \int_{\mathbf{u} \in [0,1]^m} \mathbb{C}^2(\mathbf{u}) d\mathbf{u}).$$

$\Rightarrow$ critical regions, p-values, etc.

Assumptions:

- no model assumption concerning the cdfs' F_k , $k = 1, \dots, m$
 $\Rightarrow$ any marginal cdf F_k is consistently estimated by $\hat{F}_k$, $\forall k$;
- the copula C belong to a parametric copula family
 $\mathcal{F} := \{C_{\theta_c}, \theta_c \in \Theta\}$, $\Theta \in \mathbb{R}^p$.

$\Rightarrow$ a semi-parametric model for the law of $\mathbf{X}$.

General inference strategy:

- 1 define vectors of pseudo-observations $\hat{\mathbf{U}}_i$, $i \in \{1, \dots, n\}$,

$$\hat{\mathbf{U}}_i := (\hat{U}_{i,1}, \dots, \hat{U}_{i,m}), \quad \hat{U}_{i,k} := \hat{F}_k(X_{i,k}), \quad k \in \{1, \dots, m\}.$$

- 2 apply any inference method to estimate C , “as if” $(\hat{\mathbf{U}}_i)_{i=1, \dots, n}$ were a sample of true realizations of C .

Under some conditions of regularity, this yields consistent and asymptotically estimators of θ_C .

Pros: robustness w.r.t. the specification of marginal laws.

The typical case: **Canonical Maximum Likelihood**.

Set $\hat{F}_k(t) =: F_{n,k}(t) := n^{-1} \sum_{i=1}^n 1(X_{i,k} \leq t)$, for every t and k .

Estimate θ_c by Pseudo-MLE

$$\hat{\theta}_c := \arg \min_{\theta_c} L_n^{(2)}(\theta_c) := \sum_{i=1}^n \ln c_{\theta_c}(F_{n,1}(X_{i,1}), \dots, F_{n,m}(X_{i,m})), \text{ or}$$

$$\hat{\theta}_c := \arg \min_{\theta_c} L_n^{(2)}(\theta_c) := \sum_{i=1}^n \ln c_{\theta_c}(\hat{U}_{i,1}, \dots, \hat{U}_{i,m}).$$

Ref.: Shi and Louis (1995) Biometrics, Genest, Ghoudi and Rivest (1995) Biometrika. Tsukahara (2005) Canadian J. Stat., Chen and Fan (2005), Canadian J. Stat.

Alternative inference method for copulas, that **does not depend on any assumption on margins**.

When $\theta_c \in \mathbb{R}^p$, assume there exists a set of p “dependence measures” $\kappa_1, \dots, \kappa_p$, $\kappa_k = \kappa_k(C)$, and a one-to-one mapping

$$\psi : \theta_c \mapsto (\kappa_1, \dots, \kappa_p).$$

- (i) estimate any κ_k by $\hat{\kappa}_k = \kappa_k(C_n)$ or another one consistent estimator;
- (ii) set $\hat{\theta}_c = \psi^{-1}(\hat{\kappa}_1, \dots, \hat{\kappa}_p)$.

Typically, use Kendall's tau or Spearman's rho, and use their empirical counterparts.

Example: assume C is a Clayton copula with parameter $\theta_c \in \mathbb{R}$.
The associated Kendall's tau is $\tau(C) = \theta_c/(\theta_c + 2)$.

The empirical Kendall's tau $\hat{\tau}$ consistently estimate $\tau(C)$ only with ranks (that do not depend on the margins)

Then, set $\hat{\theta}_c := 2\hat{\tau}/(1 - \hat{\tau})$.

Assumptions:

- any marginal distribution F_k , $k = 1, \dots, d$ belong to a parametric family $\mathcal{F}_k := \{F_{k,\theta_k}, \theta_k \in \Theta_k\}$, $\Theta_k \subset \mathbb{R}^{p_k}$;
- no model assumption concerning the underlying copula C .

$\Rightarrow$ another semi-parametric model for the law of $\mathbf{X}$.

A “two-stage” procedure

- (1) Estimate every marginal model, i.e. estimate θ_k for every $k \in \{1, \dots, m\} \Rightarrow \hat{\theta}_k$, a consistent estimator of θ_k .
- (2) build pseudo-observations $\hat{U}_{i,k} = F_{k,\hat{\theta}_k}(X_{i,k})$, $i \in \{1, \dots, n\}$, $k \in \{1, \dots, m\}$.
- (3) the copula C is estimated by the empirical distribution of the pseudo-sample $(\hat{U}_i)_{i=1,\dots,n}$, i.e.

$$\hat{C}_n(\mathbf{u}) := n^{-1} \sum_{i=1}^n 1(\hat{U}_i \leq \mathbf{u}), \mathbf{u} \in [0, 1]^m.$$

$\Rightarrow$ under some regularity assumptions, consistency and weak-convergence of $\sqrt{n}(\hat{C}_n - C)$... but with complex limiting laws, in general.

The limiting law of most copula estimators are complex and cannot easily be evaluated analytically.

⇒ numerical approximations by resampling techniques, in particular [the Bootstrap](#).

Bootstrap schemes provide an (almost) universal way of approximating limiting laws by simulation.

Very useful for testing purpose (calculation of p-values).

Bootstrap: the general theory

$\mathcal{X}_n$ = the available data (a sample), and $\mathbb{P}_n$ the associated empirical measure.

Ex.: $\mathcal{X}_n = (\mathbf{X}_1, \dots, \mathbf{X}_n)$ is an n -tuple of multivariate observations which may possibly be serially dependent.

$(\mathbb{D}, d)$ = a separable metric space.

Goal: approximate the law of some $\mathbb{D}$ -valued statistic $S_n = S_n(\mathcal{X}_n)$.

$\mathbb{D}$ -valued bootstrap replicates of S_n , on which inference could be based: $S_n^{(1)} = S_n^{(1)}(\mathcal{X}_n, \mathcal{W}_n^{(1)})$, $S_n^{(2)} = S_n^{(2)}(\mathcal{X}_n, \mathcal{W}_n^{(2)})$, etc., where $\mathcal{W}_n^{(j)}$, typically $\mathbb{R}^n$ -valued, are **i.i.d. weights**.

They represent additional sources of randomness s.t. $S_n^{(1)}$, $S_n^{(2)}$, etc., are independent conditionally on $\mathcal{X}_n$.

Bootstrap: the general theory

The previous setup is general enough to encompass most types of resampling procedures.

Ex.: when $\mathbb{D} = \mathbb{R}^d$, the classical empirical (multinomial) bootstrap of Efron based on **resampling with replacement** from some original i.i.d. data set $\mathcal{X}_n = (X_1, \dots, X_n)$ can be obtained by letting the $\mathcal{W}_n^{(j)} := (W_{n1}^{(j)}, \dots, W_{nn}^{(j)})$ be i.i.d. multinomially distributed with parameter $(n, 1/n, \dots, 1/n)$.

Indeed, for fixed $i \in \mathbb{N}$, the sample $\mathcal{X}_n^* = (X_1^*, \dots, X_n^*)$ constructed by including any original observation X_i exactly $W_{ni}^{(j)}$ times $\equiv$ a sample drawn with replacement from $\mathcal{X}_n$.

Many other resampling schemes: block bootstraps for time series, multiplier (or weighted, wild) bootstraps, parametric bootstrap, etc.

For all resampling scheme, the $\mathcal{W}_n^{(j)}$ can be interpreted as vectors of bootstrap weights, independent of $\mathcal{X}_n$.

Bootstrap: the general theory

Let T be an arbitrary non-empty set. $\ell^\infty(T)$ = the set of real-valued bounded functions on T equipped with the sup distance.

Here, **the situation of interest**: the $\mathbb{D}$ -valued statistic S_n is a stochastic process $\mathbb{G}_n = \mathbb{G}_n(\mathcal{X}_n)$ on T constructed from $\mathcal{X}_n$.

Assume that $\mathbb{G}_n$ may be regarded as a map from Ω into $\ell^\infty(T)$.

Suppose that, as $n \rightarrow \infty$, $\mathbb{G}_n$ converges weakly in $\ell^\infty(T)$ to some Borel measurable stochastic process $\mathbb{G}$.

$\mathbb{G}_n^{(1)} = \mathbb{G}_n^{(1)}(\mathcal{X}_n, \mathcal{W}_n^{(1)})$, $\mathbb{G}_n^{(2)} = \mathbb{G}_n^{(2)}(\mathcal{X}_n, \mathcal{W}_n^{(2)})$, etc., are bootstrap replicates of $\mathbb{G}_n$, i.e. stochastic processes on T depending on i.i.d. weights $\mathcal{W}_n^{(1)}$, $\mathcal{W}_n^{(2)}$, etc.

The sample paths of $\mathbb{G}_n^{(1)}$, $\mathbb{G}_n^{(2)}$, etc., also belong to $\ell^\infty(T)$

$BL_1(\mathbb{D}) := \{h : \mathbb{D} \rightarrow [-1, 1] \mid |h(x) - h(y)| \leq d(x, y), \forall x, y \in \mathbb{D}\}.$
(bounded Lipschitz functions defined on $\mathbb{D}$).

Proposition 7

The following assertions are equivalent:

- *As $n \rightarrow \infty$, $(\mathbb{G}_n, \mathbb{G}_n^{(1)}, \mathbb{G}_n^{(2)}) \rightsquigarrow (\mathbb{G}, \mathbb{G}^{(1)}, \mathbb{G}^{(2)})$ in $\{\ell^\infty(T)\}^3$, where the limiting random elements $\mathbb{G}$, $\mathbb{G}^{(1)}$ and $\mathbb{G}^{(2)}$ are i.i.d.*
- *For any $q \geq 2$, as $n \rightarrow \infty$, $(\mathbb{G}_n, \mathbb{G}_n^{(1)}, \mathbb{G}_n^{(2)}, \dots, \mathbb{G}_n^{(q)})$ weakly tends to $(\mathbb{G}, \mathbb{G}^{(1)}, \mathbb{G}^{(2)}, \dots, \mathbb{G}^{(q)})$ in $\{\ell^\infty(T)\}^{q+1}$, where the limiting random elements $\mathbb{G}$, $\mathbb{G}^{(1)}, \dots, \mathbb{G}^{(q)}$ are i.i.d.*
- *As $n \rightarrow \infty$,*
$$\sup_{h \in BL_1(\ell^\infty(T))} |\mathbb{E}^*[h(\mathbb{G}_n^{(1)}) \mid \mathcal{X}_n] - \mathbb{E}[h(\mathbb{G})]| \xrightarrow{P} 0.$$

The bootstrap “works”, or is “valid”:= it is safe to evaluate the limiting laws of regular test statistics by Bootstrap.

Some Bootstrap schemes

The **exchangeable bootstrap** is encompassing many bootstrap schemes: $\mathcal{W}_n = (W_{n1}, \dots, W_{nn})$ is an exchangeable, nonnegative random vector.

The weighted bootstrap empirical measure is

$$\hat{\mathbb{P}}_n := n^{-1} \sum_{i=1}^n W_{ni} \delta_{X_i}.$$

The weighted bootstrap empirical process is

$$\hat{\mathbb{G}}_n = \sqrt{n}(\hat{\mathbb{P}}_n - \overline{W}_n \mathbb{P}_n) = n^{-1/2} \sum_{i=1}^n W_{ni} (\delta_{X_i} - \mathbb{P}_n),$$

where $\overline{W}_n := n^{-1} \sum_{i=1}^n W_{ni}$.

Assume $n^{-1} \sum_{i=1}^n (W_{ni} - \overline{W}_n)^2 \xrightarrow{P} c^2 > 0$ (+others)

Examples:

- Efron nonparametric bootstrap: the vector $\mathcal{W}_n$ is drawn along a multinomial law with parameters n and probabilities $(1/n, \dots, 1/n)$. Here, $\overline{W}_n = 1$ and $c = 1$
- Wild bootstrap: the W_{ni} are i.i.d. with finite $L^{2+\epsilon}$ norms, for some $\epsilon > 0$. Now, $\overline{W}_n$ is random and $c = \text{var}(W_{1,1})$.
- Bayesian bootstrap: let Y_{ni} be i.i.d. random variables with finite $L^{2+\epsilon}$ norms, for some $\epsilon > 0$. Then, set $W_{ni} = Y_i / \overline{Y}_n$. Then, $\overline{W}_n = 1$ and $c = \sigma(Y_1)/E[Y_1]$. If the variables are exponentially distributed with mean one, this scheme is called Bayesian bootstrap.

Bootstrap and the Delta-Method: the general theory

Some general random elements $Q_n = Q_n(\mathcal{X}_n)$ and their bootstrapped versions $\hat{Q}_n := \hat{Q}_n(\mathcal{X}_n, \mathcal{W}_n)$ in a normed space, such that $\sqrt{n}(Q_n - P)$ is weakly convergent to G .

The bootstrap “works” for Q_n when $\sqrt{n}(\hat{Q}_n - Q_n)$ weakly converges given $\mathcal{X}_n$ to the same random element G , i.e.

$$\sup_{h \in BL_1(\mathbb{D})} |\mathbb{E}^* [h(\sqrt{n}(\hat{Q}_n - Q_n)) | \mathcal{X}_n] - \mathbb{E}[h(G)]|$$

Given a map $\psi : \mathbb{D} \rightarrow \mathbb{E}$, what is the behavior of $\sqrt{n}(\psi(\hat{Q}_n) - \psi(Q_n))$, given that the usual Delta-Method insures that $\sqrt{n}(\psi(Q_n) - \psi(P))$ is already weakly convergent?

To be specific, we would like to prove that the bootstrap applies for $\sqrt{n}(\psi(\hat{Q}_n) - \psi(Q_n))$ in the sense that

$$\sup_{h \in BL_1(\mathbb{E})} |\mathbb{E}^* [h(\sqrt{n}(\psi(\hat{Q}_n) - \psi(Q_n))) | \mathcal{X}_n] - \mathbb{E}[h(\psi'_P(G))]| \xrightarrow{P} 0.$$

Theorem 10

Let $\mathbb{D}$ and $\mathbb{E}$ be two normed spaces. Let $\phi : \mathbb{D}_\phi \subset \mathbb{D} \rightarrow \mathbb{E}$ be Hadamard-differentiable at P tangentially to a subspace $\mathbb{D}_0$. Let $\mathbb{Q}_n$ and $\hat{\mathbb{Q}}_n$ the random maps with values in $\mathbb{D}_\phi$, s.t. $\sqrt{n}(\mathbb{Q}_n - P) \rightsquigarrow \mathbb{G}$, where $\mathbb{G}$ is separable and takes its values in $\mathbb{D}_0$. Then $\sqrt{n}(\phi(\hat{\mathbb{Q}}_n) - \phi(\mathbb{Q}_n)) \rightsquigarrow \mathbb{G}$ under $\mathbb{Q}_n$, in the sense of (4).

Particular case: $\mathbb{Q}_n$ (resp. $\hat{\mathbb{Q}}_n$) is the empirical measure $\mathbb{P}_n$ (resp. bootstrapped empirical measure $\hat{\mathbb{P}}_n$) of $\mathcal{X}_n$, $\mathbb{G}$ is a Brownian Bridge.

General idea:

- a statistic is of the type $T_n = g(\mathbb{C}_n)$, for some continuous map $g : \ell^\infty([0, 1]^m) \rightarrow \mathbb{R}^q$
- invoke the weak convergence of the empirical copula process $\mathbb{C}_n = \sqrt{n}(C_n - C)$ (or $\overline{\mathbb{C}}_n = \sqrt{n}(\overline{C}_n - C)$), C being the true underlying copula.
- the limiting law of T_n can be approximated by the limiting law of $g(\hat{\mathbb{C}}_n)$, denoting by $\hat{\mathbb{C}}_n$ a convenient bootstrapped empirical copula process.

Dealing with an underlying parametric model, or composite assumption under a goodness-of-fit perspective, the latter principle applies most of the time, replacing $\mathbb{C}_n$ by $\sqrt{n}(C_n - C_{\hat{\theta}})$.

Some conditions of regularity are required to be able to manage $\hat{\theta}$.

$\Rightarrow$ “parametric bootstrap” (Genest & Rémillard, 2008, Annales de l’IHP).

In every case: necessary to exhibit some convenient bootstrap schemes for which the bootstrapped empirical copula process will be “valid”.

Consider a bootstrap scheme that depends on $\mathcal{W}_n$, $\mathcal{W}_n \perp\!\!\!\perp \mathcal{X}_n$.

$\Rightarrow$ the bootstrapped empirical copula processes $\hat{\mathbb{C}}_n = \hat{\mathbb{C}}_n(\mathcal{X}_n, \mathcal{W}_n)$.

We will consider independent version of the latter random element, given $\mathcal{W}_n$: $\hat{\mathbb{C}}_n^{(k)} = \hat{\mathbb{C}}_n(\mathcal{X}_n, \mathcal{W}_n^{(k)})$, $k = 1, \dots, q$, and the random vectors $\mathcal{W}_n^{(k)}$ are drawn independently.

A bootstrap scheme is valid for the empirical copula process when one the three equivalent properties is satisfied:

- As $n \rightarrow \infty$, $(\mathbb{C}_n, \hat{\mathbb{C}}_n^{(1)}, \hat{\mathbb{C}}_n^{(2)}) \rightsquigarrow (\mathbb{C}, \mathbb{C}^{(1)}, \mathbb{C}^{(2)})$ in $\{\ell^\infty(T)\}^3$, where the limiting random elements $\mathbb{C}$, $\mathbb{C}^{(1)}$ and $\mathbb{C}^{(2)}$ are i.i.d.
- For any $q \geq 2$, as $n \rightarrow \infty$, $(\mathbb{C}_n, \hat{\mathbb{C}}_n^{(1)}, \hat{\mathbb{C}}_n^{(2)}, \dots, \hat{\mathbb{C}}_n^{(q)})$ weakly tend to $(\mathbb{C}, \mathbb{C}^{(1)}, \mathbb{C}^{(2)}, \dots, \mathbb{C}^{(q)})$ in $\{\ell^\infty(T)\}^{q+1}$, where the limiting random elements $\mathbb{C}$, $\mathbb{C}^{(1)}$, $\dots$, $\mathbb{C}^{(q)}$ are i.i.d.
- $\sup_{h \in BL_1(\ell^\infty(T))} |\mathbb{E}^*[h(\hat{\mathbb{C}}_n^{(1)}) | \mathcal{X}_n] - \mathbb{E}[h(\mathbb{C})]| \xrightarrow{P} 0$.

The simplest case: the nonparametric bootstrap, with weights $\mathcal{W}_n = (w_{1n}, \dots, w_{nn})$ drawn along a multinomial distribution of size n (trials) and probabilities $(1/n, \dots, 1/n)$ (n times)

Then $\hat{\mathbb{C}}_n = \sqrt{n}(\hat{C}_n - C_n)$, and

$$\hat{C}_n(\mathbf{u}) = \frac{1}{n} \sum_{i=1}^n w_{in} 1(F_{n1}(X_{i,1}) \leq u_1, \dots, F_{n,m}(X_{i,m}) \leq u_m).$$

In practice, most results that state weak convergence of a process have a “bootstrapped” version.

Theorem 11

Assume the law F is continuous, and all the partial derivatives $\dot{C}_k$ exist and are continuous on the whole hypercube $[0, 1]^m$, $k = 1, \dots, m$. Moreover, there is a bootstrap scheme that is valid for the usual empirical process $\mathbb{F}_n$, that is associated to the initial observations $\mathcal{X}_n = (\mathbf{X}_1, \dots, \mathbf{X}_n)$. The associated bootstrapped empirical distribution is denoted by $\hat{F}_n$. Set $C_n = \phi(F_n)$, where ϕ is defined in Lemma 2. Define $\hat{C}_n := \phi(\hat{F}_n)$ and $\hat{C}_n = \sqrt{n}(\phi(\hat{F}_n) - \phi(F_n))$. Then, the latter bootstrap scheme is valid (for the empirical copula process $\mathbb{C}_n$ or $\overline{\mathbb{C}}_n$).

$\Rightarrow$ the nonparametric bootstrap and the multiplier bootstrap (Rémillard & Scaillet 2009, JMVA) are valid for copulas.

Ex.: Goodness-of-fit (cont'd)

To test $\mathcal{H}_0 : C = C_0$, consider the statistic

$$T_n^{CvM} := n \int_{[0,1]^m} (C_n - C_0)^2(\mathbf{u}) d\mathbf{u}.$$

Under $\mathcal{H}_0$, T_n^{CvM} weakly tends to $T_\infty^{CvM} := \int_{[0,1]^m} \mathbb{C}^2$.

But the law of T_n^{CvM} under $\mathcal{H}_0$ is complex $\Rightarrow$ approximation by nonparametric bootstrap.

Procedure: start from an iid sample $\mathcal{X} := (\mathbf{X}_1, \dots, \mathbf{X}_n)$.

- (i) redraw with replacement n vectors of observations $\Rightarrow$ a bootstrap sample $\mathcal{X}_n^* := (\mathbf{X}_1^*, \dots, \mathbf{X}_n^*)$;
- (ii) calculate $T_n^{CvM,*}$ based on $\mathcal{X}_n^*$;
- (iii) re-do (i) and (ii) a large number of times, say N ; we get $T_{n,1}^{CvM,*}, \dots, T_{n,N}^{CvM,*}$.
- (iv) approximate the law of T_∞^{CvM} with the latter realizations of the bootstrapped test statistic.
- (v) in particular, an approximated p-value of the test is

$$N^{-1} \sum_{k=1}^N 1(T_{n,k}^{CvM,*} > T_n^{CvM}).$$

Consider a, i.i.d. sequence of weights $(W_{n1}, \dots, W_{nn})$, with zero mean, unit variance and

$$\int_0^{+\infty} \sqrt{\mathbb{P}(|W_{in}| > x)} dx < +\infty.$$

$$\alpha'_n(\mathbf{u}) := \frac{1}{\sqrt{n}} \sum_{i=1}^n W_{in} \left\{ 1(X_{i,1} \leq F_{n,1}^-(u_1), \dots, X_{i,1} \leq F_{n,m}^-(u_m)) - C_n(\mathbf{u}) \right\}.$$

$$C'_n(\mathbf{u}) := \alpha'_n(\mathbf{u}) - \sum_{k=1}^m \hat{C}_k(\mathbf{u}) \alpha'_n(u_k, 1_{-k}),$$

where $\hat{C}_k(\mathbf{u})$ is a consistent estimator of $\dot{C}_k(\mathbf{u}) := \partial C / \partial u_k(\mathbf{u})$.

Theorem 12

If there exists a constant K s.t. $\sup_{\mathbf{u} \in [0,1]^m} |\hat{C}_k(\mathbf{u})| \leq K$ for every $k \in \{1, \dots, m\}$, and if

$$\sup_k \sup_{\mathbf{u} \in [0,1]^m, u_k \in [\delta, 1-\delta]} |\hat{C}_k(\mathbf{u}) - \dot{C}_k(\mathbf{u})| \xrightarrow{P} 0,$$

when $n \rightarrow \infty$ for all $\delta \in (0, 1/2)$, then, in $(\ell^\infty([0, 1]^m))^2$, $(\mathbb{C}_n, \mathbb{C}'_n)$ weakly tends to $(\mathbb{C}, \mathbb{C}')$, where $\mathbb{C}'$ denotes an independent copy of $\mathbb{C}$.

Ref.: Rémillard & Scaillet 2009, Testing for equality between two copulas.

Ex.: Goodness-of-fit (cont'd)

To evaluate the limiting law of T_n^{CvM} :

- (i) draw n weights $W_{in}, \dots, W_{nn}$ and obtain the associated process $\mathbb{C}'_n$;
- (ii) calculate

$$\tilde{T}_n^{CvM} := \int_{[0,1]^m} (\mathbb{C}'_n)^2$$

- (iii) given the initial sample $\mathcal{X}_n$, independently re-do (i) and (ii) a large number of times, say N ; we get $\tilde{T}_{n,1}^{CvM}, \dots, \tilde{T}_{n,N}^{CvM}$.
- (iv) approximate the law of T_∞^{CvM} with the latter realizations of the bootstrapped test statistic.
- (v) in particular, the p-value of the test is

$$N^{-1} \sum_{k=1}^N 1(\tilde{T}_{n,k}^{CvM} > T_n^{CvM}).$$

Assume the true copula is C_{θ_0} belongs to $\mathcal{C} = \{C_{\theta}, \theta \in \Theta \subset \mathbb{R}^p\}$.

The unknown parameter θ_0 is estimated by θ_n . Most typical copula parameter estimators are functions of ranks only, i.e. they depend on the pseudo-observations $\hat{\mathbf{U}}_i$, $i = 1, \dots, n$, and

$$\hat{\mathbf{U}}_i = (\hat{U}_{i1}, \dots, \hat{U}_{im}), \quad \hat{U}_{ik} = F_{nk}(X_{ik}), \quad k = 1, \dots, m.$$

C.f. the canonical maximum likelihood and the method of moments, in particular: $\theta_n = \xi(\hat{\mathbf{U}}_1, \dots, \hat{\mathbf{U}}_n)$, for some measurable map ξ .

What is the weak limit of the empirical copula process

$$\mathbb{C}_{n,p} := \sqrt{n}(C_n - C_{\theta_n}) \text{ under a parametric perspective?}$$

Here, C_n is the usual empirical copula function, and could be replaced by $\overline{C}_n$ wlog.

Bootstrap and copulas: the parametric bootstrap

- 1 given the estimated parameter θ_n , draw an iid sample $\mathcal{U}_n^* := (\mathbf{U}_1^*, \dots, \mathbf{U}_n^*)$ where every $\mathbf{U}_i^* \sim C_{\theta_n}$;
- 2 compute the associated pseudo sample $(\hat{\mathbf{U}}_1^*, \dots, \hat{\mathbf{U}}_n^*)$, based on ranks. Deduce the bootstrapped estimator $\theta_n^* := \xi(\hat{\mathbf{U}}_1^*, \dots, \hat{\mathbf{U}}_n^*)$;
- 3 denote by C_n^* the empirical copula associated to the bootstrapped sample $\mathcal{U}_n^*$:

$$C_n^*(\mathbf{u}) = F_n^*(Q_{n,1}^*(u_1), \dots, Q_{n,m}^*(u_m)), \quad \mathbf{u} \in [0, 1]^m,$$

and F_n^* (resp. $Q_{n,k}^*$) is the usual empirical cdf associated to $\mathcal{U}_n^*$ (resp. the empirical quantile function associated to $(U_{1,k}^*, \dots, U_{n,k}^*)$);

- 4 the bootstrapped copula process is $\mathbb{C}_{n,p}^* := \sqrt{n}(C_n^* - C_{\theta_n^*})$.

This procedure can be done many times, to obtain N realisations of bootstrapped quantities (samples, empirical cdfs, empirical copulas, statistics).

Ex.: Goodness-of-fit (cont'd)

To test $\mathcal{H}_0 : C = C_{\theta_0} \in \mathcal{C} := \{C_{\theta}, \theta \in \Theta\}$, against the opposite.

Let θ_n be a consistent estimator of θ_0 under $\mathcal{H}_0$.

Consider the statistic $V_n^{CvM} := n \int_{[0,1]^m} (C_n - C_{\theta_n})^2(\mathbf{u}) d\mathbf{u}$.

Under $\mathcal{H}_0$, V_n^{CvM} weakly tends to V_{∞}^{CvM} , whose law is complex $\Rightarrow$ approximation by parametric bootstrap.

Bootstrap and copulas: the parametric bootstrap

- (i) given the estimated parameter θ_n , draw an iid sample $\mathcal{U}_n^* := (\mathbf{U}_1^*, \dots, \mathbf{U}_n^*)$ where every $\mathbf{U}_i^* \sim C_{\theta_n}$;
- (ii) get the rank vectors $\mathbf{R}_1^*, \dots, \mathbf{R}_n^*$, $R_{i,k}^* := nF_{n,k}^*(U_{i,k}^*)$,

$$F_{n,k}^*(t) := \frac{1}{n} \sum_{i=1}^n 1(U_{i,k}^* \leq t), \quad k \in \{1, \dots, m\}, i \in \{1, \dots, n\}.$$

- (iii) Get a pseudo sample $(\hat{\mathbf{U}}_1^*, \dots, \hat{\mathbf{U}}_n^*)$, $\hat{U}_{i,k}^* := R_{i,k}^*/(n+1)$.
Deduce the bootstrapped estimator $\theta_n^* := \xi(\hat{\mathbf{U}}_1^*, \dots, \hat{\mathbf{U}}_n^*)$;
- (iv) denote by C_n^* the empirical copula associated to the bootstrapped sample $\mathcal{U}_n^*$: the bootstrapped copula process is $\mathbb{C}_{n,p}^* := \sqrt{n}(C_n^* - C_{\theta_n^*})$.
- (v) calculate $V_n^{\text{CvM}} := \int_{[0,1]^m} (\mathbb{C}_{n,p}^*)^2$; re-do (i) and (ii) a large number of times, say N ; we get $V_{n,1}^{\text{CvM},*}, \dots, V_{n,N}^{\text{CvM},*}$.
- (vi) approximate the law of V_∞^{CvM} with the latter realizations of the bootstrapped test statistic; in particular, the p-value of the test is $N^{-1} \sum_{k=1}^N 1(V_{n,k}^{\text{CvM},*} > V_n^{\text{CvM}})$.

Example: the Kendall tau

For any $\mathbf{U} \sim C$ in $[0, 1]^m$, define

$$(\tau + 1)/4 := \mathbb{E}[C(U_1, \dots, U_m)] = \int C(\mathbf{u}) C(d\mathbf{u}) = \int C dC.$$

We observe a sample of realizations of a m -dimensional random vector $\mathbf{X} \sim (C, F_1, \dots, F_m)$: $(\mathbf{X}_i)_{i=1, \dots, n}$.

Q: estimation of τ , without knowing C and the margins?

Sol.: build a pseudo sample $\hat{\mathbf{U}}_1, \dots, \hat{\mathbf{U}}_n$ and the empirical copula $\bar{C}_n$.

$$\begin{aligned} (\hat{\tau} + 1)/4 &:= \int \bar{C}_n(\mathbf{u}) \bar{C}_n(d\mathbf{u}) = \frac{1}{n} \sum_{i=1}^n \bar{C}_n(\hat{\mathbf{U}}_i) \\ &= \frac{1}{n} \sum_{i=1}^n \bar{C}_n(F_{n,1}(X_{i,1}), \dots, F_{n,m}(X_{i,m})). \end{aligned}$$

The map $C \mapsto \psi(C) := \int C dC$ is Hadamard differentiable at any copula C_0 in the set $\mathbb{D}$ of the cdfs' on $[0, 1]^m$: let a real sequence (t_n) , $t_n \rightarrow 0$ and (h_n) in $\mathbb{D}$.

$$\begin{aligned}\psi(C_0 + t_n h_n) - \psi(C_0) &= t_n \int \{C_0 dh_n + h_n dC_0\} + t_n^2 \int h_n dh_n \\ &= t_n \psi'_{C_0}(h_n) + O(t_n^2).\end{aligned}$$

where $\psi'_C(h) := \int h dC + \int C dh$.

Apply the Delta-Method: since $\sqrt{n}(\overline{C}_n - C) \rightsquigarrow \mathbb{C}$,

$$\sqrt{n}(\psi(\overline{C}_n) - \psi(C)) \rightsquigarrow \psi'_C(\mathbb{C}), \text{ and}$$

$$\sqrt{n}(\hat{\tau} - \tau) \xrightarrow{law} 8 \int \mathbb{C} dC \sim \mathcal{N}(0, \sigma^2).$$

Example: the Kendall's function

Similarly for the Kendall's function

$$K(t) := \mathbb{P}(C(\mathbf{U}) \leq t), \mathbf{U} \sim C, \forall t \in \mathbb{R}^+.$$

$$\hat{K}(t) := \frac{1}{n} \sum_{i=1}^n 1(\overline{C}_n(\hat{\mathbf{U}}_i) \leq t).$$

For a fixed t , it can be proved that $\sqrt{n}(\hat{K} - K)(t)$ is consistent and asymptotically normal.

Even more, the random map $t \mapsto \sqrt{n}(\hat{K}_n - K)(t)$ from $\mathbb{R}^+$ to $\mathbb{R}$ is weakly convergent (the “Kendall process”).

Ref.: Barbe, Genest, Rémillard (1996). On Kendall's process.

Example: the Pickands function A

A 2-dimensional copula C is max-stable iff there exists a convex map $A : [0, 1] \rightarrow [1/2, 1]$, $A(0) = A(1) = 1$ s.t.
 $\max(t, 1 - t) \leq A(t) \leq 1$ for every $t \in [0, 1]$, and

$$C(u, v) = \exp \left(\ln(uv) A \left(\frac{\ln v}{\ln(uv)} \right) \right).$$

Q: is it possible to estimate A non-parametrically, from an iid sample $(\mathbf{X}_i)_{i=1, \dots, n}$, $\mathbf{X} \sim (C, F, G)$?

Inference of the Pickands function

Set $U_i := F(X_{i,1})$, $V_i := G(X_{i,2})$, $i \in \{1, \dots, n\}$.

$\Rightarrow S_i := -\ln U_i$ and $T_i := -\ln V_i \sim \mathcal{E}(1)$.

$$\xi_i(t) := \min\left(\frac{S_i}{1-t}, \frac{T_i}{t}\right), \quad t \in [0, 1].$$

For every $x > 0$,

$$\begin{aligned}\mathbb{P}(\xi_i(t) > x) &= \mathbb{P}(U_i < \exp(-(1-t)x); V_i < \exp(-tx)) \\ &= C(\exp(-(1-t)x), \exp(-tx)) = \exp(-xA(t)).\end{aligned}$$

Then, $\xi_i(t)$ is exponentially distributed, with

$$\mathbb{E}[\xi_i(t)] = 1/A(t), \quad \mathbb{E}[\ln \xi_i(t)] = -\ln A(t) - \gamma,$$

where $\gamma \simeq 0.577$ is the Euler constant.

$\Rightarrow$ two estimators of A , based on pseudo observations $\hat{U}_i$, $\hat{V}_i$, $\hat{S}_i$, $\hat{T}_i$ and $\hat{\xi}(t)$:

$$\hat{A}_n^P(t) := \left\{ n^{-1} \sum_{i=1}^n \hat{\xi}_i(t) \right\}^{-1},$$

$$\ln \hat{A}^{CFG}(t) := -\gamma - \frac{1}{n} \sum_{i=1}^n \ln \hat{\xi}_i(t).$$

+ end-point corrections to satisfy the conditions of regularity of A .

Ref.: Hall & Tajvidi (2000), Caperaa, Fougères & Genest (2000), **Genest and Segers (2009)**, etc.

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